

Chapter 10

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Contents

1	Overview	2
2	Set Up	3
3	Load Data	4
3.1	Basic Data Inspection	4
4	Exclude Columns and Rows	6
4.1	Negative Indices	6
4.2	Inequality !=	7
4.3	Not !	8
4.4	Practice Relational Operators	9
5	&, , %in%	12
5.1	And &	13
5.2	Or 	15
5.3	Membership %in%	16
6	Putting &, , %in% Together	17
6.1	Grouping Parentheses	17
7	Summary	20
8	Chapter Terms	21
9	 Practice Space	23
9.1	Task 1 	24
9.2	Task 2 	24
9.3	Task 3 	25
9.4	Task 4 	26
9.5	Task 5 	26
9.6	Task 6 	28
9.7	Task 7 	28
9.8	Task 8 	29
9.9	Task 9 	30
9.10	Task 10 	30

1. Overview

In Chapter 8, you learned how bracket subsetting works: `df[rows, cols]` lets you extract and modify exactly the pieces of a data frame you need. This chapter strengthens that skill by adding three essential logical operators to your toolkit: `&` (AND), `|` (OR), and `%in%` (in set). Most real analysis questions require more than one condition—“keep rows where X is true and Y is true,” or “keep rows where A or B is true,” or “keep rows where a value is one of several allowed options.” You will practice translating those English rules into correct R code using logical tests (`>`, `<`, `==`, `!=`) combined with `&`, `|`, and `%in%`, then verify your results so you catch silent errors early—especially precedence bugs that happen when mixed operators are missing parentheses.

By the end of Chapter 10, you will be able to:

- ☐ Apply bracket subsetting `x[i, j]` to select or exclude rows and columns by position.
- ☐ Generate and interpret logical vectors using relational operators (`>`, `<`, `<=`, `>=`, `==`, `!=`) on data-frame columns accessed with `$`.
- ☐ Use logical vectors inside `[]` to filter rows based on conditions.
- ☐ Explain and use logical negation with `!` to invert tests.
- ☐ Combine multiple filtering conditions using `&` (AND), `|` (OR), and `%in%` (membership) to construct multi-criteria subsets.
- ☐ Diagnose and correct a precedence bug by adding grouping parentheses.
- ☐ Construct readable, reproducible multi-condition filters with clear comments.

Keep these goals in mind as you move through each section.

2. Set Up

To begin this guided activity, you'll want to open a new R Notebook in your course folder named `chapter10_notes.Rmd`. Update the YAML header with the title, author, and date. You can copy the YAML header from this document, and fill in your name.

```
---  
title: "Chapter 10 Notes: And, Or, and Membership"  
author: "Your Name Here"  
date: "`r Sys.Date()`"  
output: html_notebook  
---
```

3. Load Data

We will use the built-in `mtcars` dataset for this guided activity.

```
# □ Type this in your R Notebook  
data("mtcars")
```

- The `data()` function loads built-in datasets into the environment.

3.1 Basic Data Inspection

After loading data, it's good practice to inspect it.

- **Goal:** Get a quick sense of the data structure using built-in functions.

```
# □ Type this in your R Notebook  
?mtcars    # help file (opens in Help tab)  
View(mtcars) # spreadsheet view (top-left pane)
```

- The help tab shows the structure of the dataset, variable descriptions, and source information.

Let's use the subsetting skills you learned in Chapter 8 to quickly check the structure of `mtcars`.

-
- Review: Subsetting template
 - The SYNTAX: `x[i, j]` where `i` is the **row index**, `j` is the **column index**.
 - Leave ROWS blank → keep all rows.
-

- **Goal:** Get a quick sense of the data structure and content using subsetting `x[i, j]`.

```
mtcars[1:6, ]
```

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Mazda RX4	21.0	6	160	110	3.90	2.620	16.46	0	1	4	4
Mazda RX4 Wag	21.0	6	160	110	3.90	2.875	17.02	0	1	4	4
Datsun 710	22.8	4	108	93	3.85	2.320	18.61	1	1	4	1
Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44	1	0	3	1
Hornet	18.7	8	360	175	3.15	3.440	17.02	0	0	3	2
Sportabout											
Valiant	18.1	6	225	105	2.76	3.460	20.22	1	0	3	1

□ The blank after the comma means “keep all columns”. This returns the first 6 rows of `mtcars` with all columns. This subset provides the same information as `head(mtcars)`.

4. Exclude Columns and Rows

Instead of keeping rows/columns by position or condition, you can also drop them. This is often done with negative indices or with logical conditions that create TRUE/FALSE vectors.

4.1 Negative Indices

Negative indexing drops rows or columns by position. Inside the bracket subsetting operator [for vectors/matrices/data frames, a negative integer index means “leave these positions out.”

□ The SYNTAX: `x[-i, -j]`

Using negative indices drops row/column by position.

□ **Goal:** Drop the last two columns of `mtcars` (columns 8 and 9) to create a new dataframe `no_cat` that contains only numeric variables.

```
# □ Type this in your R Notebook
no_cat <- mtcars[, -c(8, 9)]
# □ Verify
head(no_cat)
```

	mpg	cyl	disp	hp	drat	wt	qsec	gear	carb
Mazda RX4	21.0	6	160	110	3.90	2.620	16.46	4	4
Mazda RX4 Wag	21.0	6	160	110	3.90	2.875	17.02	4	4
Datsun 710	22.8	4	108	93	3.85	2.320	18.61	4	1
Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44	3	1
Hornet Sportabout	18.7	8	360	175	3.15	3.440	17.02	3	2
Valiant	18.1	6	225	105	2.76	3.460	20.22	3	1

□ The `-c(8, 9)` is the integer vector containing -8 and -9. Negative column indices mean: drop column 8 and drop column 9. The blank before the comma, `i` is empty → all rows. For `mtcars`, columns 8 and 9 are `vs` and `am`, respectively. So `mtcars[, -c(8, 9)]` returns all rows and all columns except `vs` and `am`.

□ **Tip:** With integer indexing, you generally use all positive indices (keep these) or all negative indices (drop these). Mixing them triggers the classic error (“only 0’s may be mixed with negative subscripts”).

The use of the negative sign `-` is one way to subset by excluding columns. However, the most powerful way to subset dataframes is with logical operators that create TRUE/FALSE vectors.

4.2 Inequality !=

Conditional subsetting with logical (relational) operators is more flexible than negative indexing because you can specify conditions on the values in the data frame, not just positions. Recall from Chapter 8 that logical tests create TRUE/FALSE vectors used inside of `[]`.

Not Equal To (`!=`) is a relational operator that compares values and returns a logical vector of TRUE/FALSE results. When the element value is not equal to the specified value, the result is TRUE. When the element value is equal to the specified value, the result is FALSE. The result is the opposite of the `==` operator, which tests for equality.

□ **Goal:** Use `!=` to create a logical vector that is TRUE where the number of cylinders (`cyl`) is NOT equal to 4.

```
# □ Type this in your R Notebook
mtcars$cyl != 4
```

```
## [1] TRUE TRUE FALSE TRUE TRUE TRUE TRUE FALSE FALSE TRUE TRUE TRUE
## [13] TRUE TRUE TRUE TRUE TRUE FALSE FALSE FALSE FALSE TRUE TRUE TRUE
## [25] TRUE FALSE FALSE FALSE TRUE TRUE TRUE FALSE
```

□ This does not return a subset. It produces a **logical vector** you can use for subsetting, typically rows. In this example, the code returns a logical vector with TRUE where the number of gears is anything but 4 and FALSE otherwise.

□ **Review:** Accessing Columns with `$`

- `$` extracts a single column as a vector.
- □ The SYNTAX: `dataframe$column_name`

□ **Goal:** Use `$` to access the `cyl` column of `mtcars` as a vector.

```
# □ Review
mtcars$cyl
```

```
## [1] 6 6 4 6 8 6 8 4 4 6 6 8 8 8 8 8 4 4 4 4 8 8 8 8 4 4 4 8 6 8 4
```

□ The `$` operator is used to access ONE specific column in a data frame by name. The column name is `cyl` and it tells the number of cylinders 4, 6, or 8 for each car. The vector lists the values in `cyl` in row order. The TRUE and FALSE vector returned from `mtcars$cyl != 4` returned TRUE when the value did not equal 4.

For example, the first value 6 is TRUE because 6 is not equal to 4. The second value 6 is also TRUE because 6 is not equal to 4. The third value 4 is FALSE because 4 is equal to 4.

□ **Goal:** Use the logical test `!= 4` inside `[]` to keep only rows where `cyl` is NOT equal to 4.

```
# □ Type this in your R Notebook
not4 <- mtcars[mtcars$cyl != 4, ]
# □ Verify
head(not4)
```

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Mazda RX4	21.0	6	160	110	3.90	2.620	16.46	0	1	4	4
Mazda RX4 Wag	21.0	6	160	110	3.90	2.875	17.02	0	1	4	4
Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44	1	0	3	1
Hornet	18.7	8	360	175	3.15	3.440	17.02	0	0	3	2
Sportabout											
Valiant	18.1	6	225	105	2.76	3.460	20.22	1	0	3	1
Duster 360	14.3	8	360	245	3.21	3.570	15.84	0	0	3	4

□ This returns **a subset** of `mtcars` with all rows where the number of cylinders is anything but 4 and all columns. When the logical test returned `TRUE`, the row is selected. When the logical test returned `FALSE` the row is removed from the output.

4.3 Not !

The **Not** operator `!` negates a logical value. `!TRUE` becomes `FALSE`, and `!FALSE` becomes `TRUE`. It can be used to flip the result of a logical test.

□ **Goal:** Use `!` to create a logical vector that is `TRUE` where the number of cylinders (`cyl`) is NOT equal to 4.

```
# □ Type this in your R Notebook
not4_bang <- mtcars[!(mtcars$cyl == 4), ]
head(not4_bang)
```

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Mazda RX4	21.0	6	160	110	3.90	2.620	16.46	0	1	4	4
Mazda RX4 Wag	21.0	6	160	110	3.90	2.875	17.02	0	1	4	4
Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44	1	0	3	1
Hornet	18.7	8	360	175	3.15	3.440	17.02	0	0	3	2
Sportabout											
Valiant	18.1	6	225	105	2.76	3.460	20.22	1	0	3	1
Duster 360	14.3	8	360	245	3.21	3.570	15.84	0	0	3	4

□ $!(x == 4)$ and $x != 4$ mean the same thing! $!=$ is a direct “not equal” comparison. $!$ is more general. It negates a logical test. Here, $!(mtcars$cyl == 4)$ reads as “NOT (cyl equals 4)”, which is equivalent to “cyl not equal to 4”.

□ **Goal:** Verify that the $!$ version of the subset matches the $!=$ version.

```
# □ Verify it matches the != version  
head(not4_bang == not4)
```

```
##           mpg cyl disp  hp drat   wt  qsec vs   am gear carb  
## Mazda RX4      TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE  
## Mazda RX4 Wag  TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE  
## Hornet 4 Drive  TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE  
## Hornet Sportabout TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE  
## Valiant        TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE  
## Duster 360     TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
```

□ The two subsets are the same.

The $!$ version is more flexible because you can negate *any* logical test, not just equality.

In summary, **negative indices** - excludes rows/columns by position only. The **inequality** $!=$ operator compares values and returns TRUE/FALSE (great for filtering rows). The **not** $!$ operator negates any logical test. **Not** is great for more complex logical tests such as element-by-element or membership logical operators covered later in this chapter.

4.4 Practice Relational Operators

Before we dive into complex logical tests, let's make sure we understand how we can combine these relational operators together. So far in this chapter, we have used the logical operators $==$, $!=$, and $!$. Let's practice using the relational operators in combination with subsetting.

4.4.1 Example 1

The **Less Than or Equal To** ($<=$) operator returns TRUE where the left side is less than or equal to the right side, and FALSE otherwise.

□ **Goal:** Answer the question, “How many cars in the dataset have $mpg \leq 15$?” Use a logical test to count how many cars have miles per gallon less than or equal to 15.

```
# □ Type this in your R Notebook in a code chunk  
sum(mtcars$mpg <= 15)
```

```
## [1] 6
```

□ The logical test `<= 15` evaluates each row value in the column `mpg` in the `mtcars` dataset (`mtcars$mpg`). If the car in a row has a value less than or equal to 15, the result is `TRUE`, otherwise `FALSE`. In R, `TRUE` is treated as 1 and `FALSE` as 0. The function `sum()` adds all the `TRUE` values. There are 6 cars in the `mtcars` dataset with mile per gallon less than or equal to 15.

4.4.2 Example 2

The **Greater Than or Equal To** (`>=`) operator returns `TRUE` where the left side is greater than or equal to the right side, and `FALSE` otherwise. We can combine this operator with negative indexing to create a subset of the data frame.

□ **Goal:** Create a subset of `mtcars` that contains rows with `hp >= 100`, and drops columns 3, 5, and 8:9 by position.

```
# □ Type this in your R Notebook in a code chunk
selection <- mtcars[mtcars$hp >= 100, -c(3, 5, 8:9)]
# □ Verify
head(selection)
```

	mpg	cyl	hp	wt	qsec	gear	carb
Mazda RX4	21.0	6	110	2.620	16.46	4	4
Mazda RX4 Wag	21.0	6	110	2.875	17.02	4	4
Hornet 4 Drive	21.4	6	110	3.215	19.44	3	1
Hornet Sportabout	18.7	8	175	3.440	17.02	3	2
Valiant	18.1	6	105	3.460	20.22	3	1
Duster 360	14.3	8	245	3.570	15.84	3	4

□ `mtcars$hp >= 100` keeps only rows where horsepower is at least 100. The `-c(3, 5, 8:9)` part drops columns 3, 5, 8, and 9 from the result (i.e., it removes those column positions).

□ **Tip:** A common mistake is to forget which side the `>`, `<` goes to the `=`. Remember that the `=` is always on the *right side* of the operator.

□ `>=`

□ `=>`

4.4.3 Example 3

We can also verify that we got the expected number of columns in our subset by passing the stored object names through functions (like `ncol()` counts columns) and then compare the results with `==`.

□ **Goal:** Verify that `selection` has the expected number of columns (i.e., 7 columns, since we dropped 4 from the original 11).

```
# □ Verify  
ncol(selection) == ncol(mtcars) - length(c(3, 5, 8:9))
```

```
## [1] TRUE
```

□ This checks whether the subset has **exactly the expected number of columns** after dropping columns 3, 5, 8, and 9.

- `ncol(selection)` counts the number of columns in the selection data frame.
- `ncol(mtcars)` counts the number of columns in the original `mtcars` dataset (which is 11).
- `length(c(3, 5, 8:9))` counts how many columns we dropped (which is 4). *Note that we could have written it as `c(3, 5, 8, 9)`.*
- The `ncol(mtcars)` (11) minus `length(c(3, 5, 8:9))` (4) gives us the expected number of columns in selection (which is 7).

If it returns `TRUE`, we removed the right columns (7 remaining); if it returns `FALSE`, something went wrong (e.g., wrong column positions, accidental extra drops).

□ **Tip:** For a data analyst, doing quick verification steps to prevent silent errors before modeling or reporting. We need to ensure our dataset structure matches our assumptions so downstream results aren't based on the wrong variables.

The next section introduces three new logical operators that help with more complex conditions: `&` (and), `|` (or), and `%in%` (in set).

5. **&**, **|**, **%in%**

Up to this point, you have used single logical tests inside brackets to pick rows (for example, `mtcars[mtcars$mpg <= 15, ]`). That works when your filter has one requirement.

What if we have multiple requirements?

- “I want cars that are automatic **AND** have $\text{mpg} \leq 22$.”
- “I want cars with 4 **OR** 8 cylinders.”
- “I want cylinders in the set $\{4, 6\}$.”

This section adds three new primary operators to our R toolkit.

- **&** means **AND** and checks that both conditions are true.
- **|** means **OR** and check if at least one condition is true
- **%in%** means membership (**in set**) and checks if a value is within a set

These operators allow us to combine multiple conditions for more complex subsetting.

Here is a summary table showing primary logical operators.

Operator	Meaning	Example
<code>></code>	greater than	<code>mpg > 20</code>
<code><</code>	less than	<code>mpg < 15</code>
<code>==</code>	equal to	<code>am == 1</code>
<code>>=</code>	greater <i>or</i> equal	<code>mpg >= 25</code>
<code><=</code>	less <i>or</i> equal	<code>mpg <= 15</code>
<code>!</code>	not	<code>!(gear %in% c(3,4))</code>
<code>!=</code>	not equal to	<code>gear != 4</code>
<code>&</code>	and	<code>x > 2 & y == "a"</code>
<code> </code>	or	<code>cyl == 4 cyl == 8</code>
<code>%in%</code>	in set	<code>region %in% c("West", "South")</code>

Let's explore each of these three new operators in more detail. In this section, we'll build multiple logical tests on columns, combine tests with **&** and **|** to form a single condition, and use **%in%** to match one of many allowed values.

5.1 And &

Suppose we have two events: A and B. The **And** operator & returns TRUE only if **both** event A and event B are **true**. It returns FALSE if either event A or event B (or both) are false. The “and” is used to combine two or more conditions where all conditions must be true.



□ **Goal:** filter mtcars to keep cars that are both automatic (am == 0) *and* have the average miles per gallon is less than or equal to 22.

Step 1: Create two separate logical vectors for each condition.

```
# □ Type this in your R Notebook in a code chunk
is_auto <- mtcars$am == 0
mpg_22_or_less <- mtcars$mpg <= 22

# □ Verify: look at the first few TRUE/FALSE values
head(is_auto)
```

```
## [1] FALSE FALSE FALSE TRUE TRUE TRUE
```

```
head(mpg_22_or_less)
```

```
## [1] TRUE TRUE FALSE TRUE TRUE TRUE
```

□ Two logical vectors are created. is_auto is TRUE where the car has automatic transmission (am == 0) and FALSE where it has manual transmission (am == 1). mpg_22_or_less is TRUE where the miles per gallon is less than or equal to 22, and FALSE otherwise.

Step 2: Combine the two logical vectors with & to create a single condition that is TRUE only where both conditions are met.

```
# □ Combine conditions (row-by-row)
keep_rows <- is_auto & mpg_22_or_less

# □ Verify
head(keep_rows)
```

```
## [1] FALSE FALSE FALSE TRUE TRUE TRUE
```

```
sum(keep_rows) # how many rows will be kept?
```

```
## [1] 17
```

□ The `keep_rows` vector is `TRUE` only where both `is_auto` and `mpg_22_or_less` are `TRUE`. The `sum(keep_rows)` counts how many rows satisfy both conditions.

Step 3: Use the combined logical vector to subset the data frame.

```
# □ Subset the data
```

```
auto_mpg_22_or_less <- mtcars[keep_rows, ]
```

```
# □ Verify
```

```
head(auto_mpg_22_or_less)
```

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215	19.44	1	0	3	1
Hornet	18.7	8	360.0	175	3.15	3.440	17.02	0	0	3	2
Sportabout											
Valiant	18.1	6	225.0	105	2.76	3.460	20.22	1	0	3	1
Duster 360	14.3	8	360.0	245	3.21	3.570	15.84	0	0	3	4
Merc 280	19.2	6	167.6	123	3.92	3.440	18.30	1	0	4	4
Merc 280C	17.8	6	167.6	123	3.92	3.440	18.90	1	0	4	4

□ The `auto_mpg_22_or_less` is a subset data frame of `mtcars` that contains only rows where the car is automatic and has miles per gallon less than or equal to 22.

Alternatively, you can combine the conditions directly inside the subset without creating intermediate logical vectors.

```
# □ Type this in your R Notebook in a code chunk
```

```
auto_mpg_22_or_less <- mtcars[ mtcars$am == 0 & mtcars$mpg <= 22 , ]
```

□ Use `&` when you want rows that satisfy all requirements at the same time. Both `mtcars$am == 0` and `mtcars$mpg <= 22` must meet the condition to be `TRUE`.

```
# □ Verify
```

```
head(auto_mpg_22_or_less)
```

```
# □ Verify
```

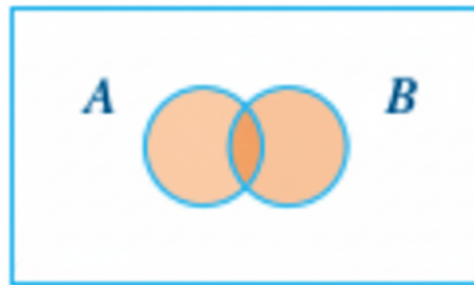
```
nrow(auto_mpg_22_or_less)
```

□ `&` is the correct operator for the creation of `auto_mpg_22_or_less` because we want cars that are *both* automatic and have mpg less than or equal to 22. `&` is strict. It requires *everything* to be true.

□ **Tip:** Students often write one condition correctly and “half-write” the other, like: `mtcars$am == 0 & mpg <= 22` ← wrong because `mpg` is not defined by itself here. Both sides of `&` must be complete logical tests.

5.2 Or |

Suppose we have two events: A and B. The **Or** logical operator `|` returns TRUE if either **event A** is true, **event B** is true, **or both** events are **true**. It only returns FALSE if both events are false. “Or” is inclusive and is used to combine two or more conditions where at least one condition must be true.



Use `|` when you want rows that satisfy at least one requirement. The `|` is located above the Enter key on your keyboard. It is the vertical bar and shares the key with the backslash `\`. Click and hold the Shift key to type `|`.

□ **Goal:** keep rows (cars) that with either 4 *or* 8 cylinders.

□ *Type this in your R Notebook in a code chunk*

```
four_or_eight <- mtcars[mtcars$cyl == 4 | mtcars$cyl == 8, ]
```

□ *Verify*

```
head(four_or_eight)
```

□ *Verify*

```
unique(four_or_eight$cyl)
```

□ The Or `|` operator checks element-by-element to return only cars that have either 4 or 8 cylinders.

□ **Tip:** Sometimes students mix up when to use `|` where they meant `&` (or vice versa). Try saying the filter outloud before coding.

- “I want rows that are both A **and** B” are TRUE at the same time. □ use `&`
- “I want rows where at least one (A **or** B)” is TRUE. □ use `|`

5.3 Membership %in%

OR becomes annoying when you have many options. The **in set** %in% operator tests if values belong to a set (vector). It is a shorthand for multiple OR | conditions. This is “the scalable OR operator” when you have many values to match.

The %in% in x %in% c(a, b, c) means “For each value in x, is it one of these allowed values?” It returns a logical vector (TRUE/FALSE) the same length as x.

□ **Goal:** Same task as the | example but this time with %in%.

```
# □ Type this in your R Notebook in a code chunk
four_or_eight_2 <- mtcars[mtcars$cyl %in% c(4, 8), ]
```

```
# □ Verify: same number of rows as the | version
nrow(four_or_eight) == nrow(four_or_eight_2)
```

□ When matching many categories, %in% is easier than chaining many | statements. %in% prevents repetitive OR chains. The %in% stays readable when the list is long.

□ **Goal:** Use %in% to filter mtcars for cars with 4, 6, or 8 carburetors.

```
# □ Tesype this in your R Notebook in a code chunk
carb_4_6_8 <- mtcars[mtcars$carb %in% c(3, 4, 6, 8), ]
```

```
# □ Verify
head(carb_4_6_8)
```

□ If we had used | instead of %in%, we would have had to write mtcars\$carb == 3 | mtcars\$carb == 4 | mtcars\$carb == 6 | mtcars\$carb == 8, which is much more verbose and harder to read.

□ **Tip:** A common mistake with %in% is mixing the order.

- c(4, 8) %in% mtcars\$cyl is □ invalid code, wrong meaning for filtering rows
- use the x %in% c(a, b, c) format for subsetting rows. The left side is the column you are testing, and the right side is the set of allowed values.

Now that we have these three tools for building row filters, we can combine them to create more complex conditions.

6. Putting `&`, `|`, `%in%` Together

In this section, you'll practice combining `&`, `|`, `%in%` into realistic filters. The hard part is not typing the operators—it's making sure your code matches the sentence you think you wrote.

6.1 Grouping Parentheses

When we combine multiple conditions with `&` and `|` or `%in%`, we need to make sure the code matches the sentence we think it does. Use **grouping parentheses** when mixing `&` and `|` to make intent explicit. Without grouping parentheses, we can trigger a **precedence bug**, where R groups a mixed `&|` filter by its precedence rules (treating `a & b | c` as `(a & b) | c`) instead of the logic we intended (`a & (b | c)`). The **grouping parentheses** overrides the operator precedence.

Add parentheses to show what the OR part is.

Let's see how the parentheses changes which rows are selected.

□ **Goal:** Keep automatic transmission cars (`0 = automatic`) and either are very efficient (`mpg ≥ 25`) or very light (`wt < 2.5`, i.e., `< 2500 lb`).

We'll write this with and without parentheses to see the difference.

```
# □ Type this in your R Notebook in a code chunk
maybe_wrong <- mtcars[ mtcars$am == 0 & mtcars$mpg >= 25 | mtcars$wt < 2.5 , ]
head(maybe_wrong)
```

	mpg	cyl	displacement	hp	drat	wt	qsec	vs	am	gear	carb
Datsun 710	22.8	4	108.0	93	3.85	2.320	18.61	1	1	4	1
Fiat 128	32.4	4	78.7	66	4.08	2.200	19.47	1	1	4	1
Honda Civic	30.4	4	75.7	52	4.93	1.615	18.52	1	1	4	2
Toyota Corolla	33.9	4	71.1	65	4.22	1.835	19.90	1	1	4	1
Toyota Corona	21.5	4	120.1	97	3.70	2.465	20.01	1	0	3	1
Fiat X1-9	27.3	4	79.0	66	4.08	1.935	18.90	1	1	4	1

□ The statement was evaluated like this: `(mtcars$am == 0 & mtcars$mpg >= 25) OR mtcars$wt < 2.5`. This means that we kept cars that are both automatic and very efficient, or we kept cars that are very light. This is not our defined goal.

```
# □ Type this in your R Notebook in a code chunk
intended <- mtcars[ mtcars$am == 0 & (mtcars$mpg >= 25 | mtcars$wt < 2.5) , ]
head(intended)
```

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Toyota	21.5	4	120.1	97	3.7	2.465	20.01	1	0	3	1
Corona											

□ The statement was evaluated like this: `mtcars$am == 0 AND (mtcars$mpg >= 25 | mtcars$wt < 2.5)`. This means that we kept cars that were automatic and either are very efficient or very light. That's our goal!

□ Compare the number of rows in `maybe_wrong` to `intended`.

```
# □ Verify
nrow(maybe_wrong) == nrow(intended)
```

□ The parentheses matter when mixing `&` and `|` because parentheses clarify the order of operations. Without parentheses, the expression is evaluated left to right, which can lead to unintended results. With parentheses, we ensure that the conditions within them are evaluated first.

6.1.1 Multi-Condition Filter

The previous example was about controlling meaning. Now you'll apply that idea to a filter that looks like something you'd actually do in analysis: restrict the dataset using several criteria at once.

□ **Goal:** Pick cars that weigh $< 3,000$ lb, have horse power between 110 and 180, and cylinders $\{4,6\}$.

In mtcars, weight is in units of 1000 lbs, so we want `wt < 3`.

□ **Tip:** Before coding, translate the goal into a single sentence that matches the logic you want. "I want cars that are light ($wt < 3$) **AND** have mid-range horsepower ($110 \leq hp \leq 180$) **AND** have either 4 OR 6 cylinders."

```
# □ Type this in your R Notebook in a code chunk
light_midpower <- mtcars[
  mtcars$wt < 3 &
  mtcars$hp >= 110 & mtcars$hp <= 180 &
  mtcars$cyl %in% c(4, 6), ]
```

□ We wrote each condition on a separate line for readability, but they are combined into a single filter. By having one condition per line, we can visually identify the logical arguments. When you click "Enter" (Return), R automatically indents.

```
# □ Verify
head(light_midpower)
```

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.46	0	1	4	4
Mazda RX4	21.0	6	160.0	110	3.90	2.875	17.02	0	1	4	4
Wag											
Lotus Europa	30.4	4	95.1	113	3.77	1.513	16.90	1	1	5	2
Ferrari Dino	19.7	6	145.0	175	3.62	2.770	15.50	0	1	5	6

```
# □ Verify
range(light_midpower$hp)
```

```
## [1] 110 175
```

The code selected rows with horsepower between 110 and 180.

```
# □ Verify
unique(light_midpower$cyl)
```

```
## [1] 6 4
```

The code selected rows with 4 or 6 cylinders.

```
# □ Verify
all(light_midpower$wt < 3)
```

```
## [1] TRUE
```

The code selected rows with weight under 3 (i.e., under 3000 lbs).

By this point, you should be able to translate a real-world rule into a logical filter, use parentheses to prevent a precedence bug, and confirm that your subset matches your intent with quick verification checks. Next, we'll step back and summarize the core subsetting and logic skills you practiced in this chapter and why they matter for reliable analysis.

7. Summary

You learned how to subset data frames more precisely by building logical tests and using them inside `df[rows, cols]`. You practiced excluding columns with negative indexing, filtering rows with relational operators like `!=`, and negating logical tests with `!`. You then added three high-impact operators: `&` to require multiple conditions at the same time, `|` to accept at least one condition, and `%in%` to match values that belong to a set without writing long OR chains. Most importantly, you learned to prevent a precedence bug by using grouping parentheses whenever you mix `&` and `|`, so the code matches the sentence you intended. If you can state your rule clearly, write the logical filter cleanly, and verify the output with quick checks (`nrow()`, `unique()`, `range()`, `all()`), you can trust that the subset you pass into later analysis is the right one.

8. Chapter Terms

And (&): An element-by-element logical operator that returns TRUE only where both conditions are TRUE; otherwise it returns FALSE (or NA when the result can't be determined because of missing values).

Conditional Subsetting (Logical Subsetting): Filtering an object by using a logical condition inside `[` (e.g., `df[df$cyl != 4, ]`), where TRUE keeps an element/row and FALSE drops it. This contrasts with negative indexing, which excludes by position rather than by a condition.

Equal To (==): A relational operator that tests equality and returns TRUE where values are equal and FALSE where they are not (element-by-element for vectors).

Greater Than (>): A relational operator that compares two values (or vectors) and returns a logical vector that is TRUE where the left side is greater than the right side, and FALSE otherwise (with NA if the comparison is unknown).

Greater Than or Equal To (>=): A relational operator that returns TRUE where the left side is greater than or equal to the right side, and FALSE otherwise.

Grouping Parentheses: Parentheses `()` used to explicitly group parts of an expression so R evaluates them together, in the order you intend, rather than relying on operator precedence and associativity rules. This is especially important when mixing logical operators like `&` and `|`, where missing parentheses can change the meaning of your filter.

In Set (%in%): A membership operator that returns a logical vector indicating whether each element on the left appears anywhere in the vector on the right (e.g., `region %in% c("West", "South")`).

Inequality Operator (!=): A relational (comparison) operator that tests whether values are not equal and returns a logical vector (TRUE/FALSE) element-by-element (e.g., `x != 4`). By itself it doesn't drop anything—it creates a condition you can use for subsetting.

Less Than (<): A relational operator that returns TRUE where the left side is less than the right side, and FALSE otherwise (vectorized across elements).

Less Than or Equal To (<=): A relational operator that returns TRUE where the left side is less than or equal to the right side, and FALSE otherwise.

Negative Indexing: Using negative integers inside the single bracket operator `[` to exclude elements (or rows/columns) by position (e.g., `x[-2]` drops the 2nd element; `df[, -c(8, 9)]` drops columns 8 and 9). In base R, negative and positive indices can't be mixed in the same subset (except 0).

Not (!): A logical operator that negates a logical value: `!TRUE` becomes FALSE, and `!FALSE` becomes TRUE (and `!NA` remains NA).

Not Equal To (!=): A relational operator that returns TRUE where values differ and FALSE where they match (vectorized across elements).

Or (|): An element-by-element logical operator that returns TRUE where either condition is TRUE; otherwise it returns FALSE (or NA when the result is ambiguous due to missing values).

Precedence Bug: An error where a mixed-operator expression is grouped and evaluated differently than intended because R applies operator precedence rules when parentheses are missing. In R, & binds more tightly than |, so `a & b | c` is interpreted as `(a & b) | c` unless you add parentheses to force `a & (b | c)`.

9. ☒ Practice Space

In this Practice Space, you will practice writing logical filters for data frames using bracket subsetting (`df[rows, cols]`) and the operators you learned in this chapter:

- relational operators (`<`, `<=`, `>=`, `==`, `!=`)
- logical operators (`&`, `|`)
- membership (`%in%`)
- indexing with `which()`

You'll apply these skills to an **NGO health pilot** scenario using the `gapminder` dataset to narrow thousands of rows down to a short, defensible shortlist.

How to complete and submit

1. Open a new R Notebook and save it in your course folder as: `chapter10_practice.Rmd`. Edit the YAML so that it includes the title “Chapter 10 Practice”, your name as the author, and the date. Remove the default content in the R Notebook. Create heading for each task.
2. Work through each task in order. Replace every `___` with code or a short written answer.
3. Run each completed chunk as you go. After each task, check that new objects (like `focus_pool`, `brief_sheet`, `needs_screen`) appear in the Environment.
4. Knit your R Notebook to HTML. Submit the HTML file (not the `.Rmd`).

NGO Health Pilot ☐☐

Imagine your nonprofit received 100 vaccine coolers and one charter flight. You must assemble a shortlist of countries for a health-equity pilot within the next few weeks. Use `dslabs::gapminder` to (a) focus on specific regions, (b) filter by need indicators, and (c) apply a simple feasibility window based on population size.

☐ Setup & Orientation

☐ **Goal:** Load the data, read the help page, and become familiar with the `gapminder` dataset.

We installed the package `dslabs` in Chapter 7. `dslabs` has curated datasets from Harvard/X data-science labs for data science education.

```
# □ Type this in your R Notebook in a code chunk

# install.packages("dslabs") # uncomment if you have not yet installed dslabs

# Load the package and data
library(dslabs) # use library() to load the package

gapminder <- dslabs::gapminder # load the dataset "gapminder"

?gapminder # pull up gapminder in the help window
```

```
colnames(gapminder) # use colnames() to list names of the columns in gapminder
```

□ How many rows are in gapminder?

9.1 Task 1

□ Pick a Year

□ Identify the years present in the data, store the 2010 as target_year.

Step 1 Use unique() to print the available year in gapminder.

```
# □ Type this in your R Notebook, Fill in the blanks
# TODO 1 □ A: list years
unique(gapminder$____) # see unique `year` in `gapminder`
```

Step 2 Store 2010 as target_year.

```
# □ Type this in your R Notebook, Fill in the blanks
# TODO 1 □ B: Select the 2010 year
target_year <- ____ # must be a single year of gapminder
```

□ We use 2010 because it has the most complete data

9.2 Task 2

□ Pick your regions

□ Identify the exact region names; then pick (your choice) your two target regions.

Step 1 Identify the unique region options in gapminder.

```
# ☐ Type this in your R Notebook, Fill in the blanks
# TODO 2-A: Show all region options
____(____$____) # Fill in: function, dataset, and column name
```

Use this output to select two regions. Copy exact names.

Step 2

```
# ☐ Type this in your R Notebook, Fill in the blanks
# TODO 2-B: store two regions EXACTLY as shown above
region_1 <- "____"
region_2 <- "____"
```

☐ I selected ____ and ____ as my target regions because _____.

9.3 Task 3

☐ Focus the Pool

☐ Create focus_pool that keeps rows from the target_year AND restricts to two regions you want to prioritize.

Step 1 Build the logical tests for year and region separately.

```
# ☐ Type this in your R Notebook, Fill in the blanks
# TODO 3-A: build the logical tests (two separate vectors)

is_year <- gapminder$year ____ target_year # test for equality

is_region <- _____$region ____ c(region_1, region_2) # test for membership in the set
```

Step 2 Combine the two logical vectors with & to create a single filter.

```
# ☐ Type this in your R Notebook, Fill in the blanks
# TODO 3-B: combine

keep <- _____ - _____ # is_year and is_region with AND
```

Step 3 Use the combined logical vector to subset gapminder and create focus_pool.

```
# ☐ Type this in your R Notebook, Fill in the blanks
# TODO 3-C: subset

focus_pool <- gapminder[____, ____] # subset rows with keep, keep all columns
```

☐ Describe what %in% and & are doing here, and why you used \$ with column names.

EXPLANATION: “____”

Compare the number of rows in gapminder with the number of rows in focus_pool

```
# ☐ Type this in your R Notebook, Fill in the blanks
# TODO 2-D: Compare number of rows in focus_pool with the original dataset
____(focus_pool) — ____ (gapminder)
```

☐ How do the number of rows in focus_pool compare to gapminder?

9.4 Task 4

☐ Negative Indices

Create a simpler briefing table brief_sheet by dropping exactly columns by position (not by name).

Use negative indexing to drop the columns 6, 7, 8, and 9 from focus_pool.

```
# ☐ Type this in your R Notebook, Fill in the blanks
brief_sheet <- focus_pool[ , ____ (6, 7, 8, 9) ] # use negative c()
# ☐ Verify
names(_____)
```

☐ Explain how negative indices work for column selection.

EXPLANATION: “____”

The brief_sheet is a simpler table with only the columns that have data. This makes it easier to inspect the indicators we care about in the next task.

9.5 Task 5

☐ In task 6, you will identify countries that need the health pilot the most. You will use three indicators as filters: life_expectancy, infant_mortality, and fertility. Before you set thresholds for these indicators, you should inspect the range of values to make informed decisions.

☐ Inspect before choosing thresholds

9.5.1 Life Expectancy (a)

Step 1 Use range() to inspect the range of values in life_expectancy

```
# ☐ Type this in your R Notebook, Fill in the blanks
# The value `a` in Task 6 will be based on this range and count.
range(brief_sheet$life_expectancy, na.rm = _____)
```

☐ The range of life expectancy in `brief_sheet` is from ____ to ____ years.

Step 2 Use `sum()` to count how many countries have life expectancy above a certain threshold (e.g., 70 years). Select a value within the range you found in Step 1.

```
# ☐ Type this in your R Notebook, Fill in the blanks  
# The value `a` in Task 6 will be based on this range and count.  
sum(brief_sheet$life_expectancy <= ____)
```

☐ There are ____ countries with life expectancy above ____ years.

9.5.2 Infant Mortality (b)

Step 3 Use `range()` to inspect the range of values in `infant_mortality`

```
# ☐ Type this in your R Notebook, Fill in the blanks  
# The value `b` in Task 6 will be based on this range and count.  
range(brief_sheet$, na.rm = TRUE)
```

☐ The range of infant mortality in `brief_sheet` is from ____ to ____ deaths per 1000 live births.

Step 4 Use `sum()` to count how many countries have infant mortality above a certain threshold (e.g., 35 deaths per 1000 live births). Select a value within the range you found in Step 3.

```
# ☐ Type this in your R Notebook, Fill in the blanks  
# The value `b` in Task 6 will be based on this range and count.  
sum(brief_sheet$infant_mortality >= ____)
```

☐ There are ____ countries with infant mortality above ____ deaths per 1000 live births.

9.5.3 Fertility (c)

Step 5 Use `range()` to inspect the range of values in `fertility`

```
# ☐ Type this in your R Notebook, Fill in the blanks  
# The value `c` in Task 6 will be based on this range and count.  
range(____$, na.rm = TRUE)
```

☐ The range of fertility in `brief_sheet` is from ____ to ____

Step 6 Use `sum()` to count how many countries have fertility above a certain threshold (e.g., 2 children per woman). Select a value within the range you found in Step 5.

```
# ☐ Type this in your R Notebook, Fill in the blanks  
# The value `c` in Task 6 will be based on this range and count.  
sum(brief_sheet$fertility >= ____)
```

☐ There are ____ countries with fertility above ____ children per woman.

9.6 Task 6 ☒

- Now that you have inspected the ranges and counts for each indicator, you can set thresholds for each one to identify countries that need the health pilot the most.
- Use what you learned from Task 5 to determine appropriate cutoff values for life_expectancy (a), infant_mortality (b), and fertility (c).

Note: High fertility is typically used as a proxy for higher maternal/child health burden and a greater strain on the health systems. More developed countries tend to have lower fertility rates.

□ *Type this in your R Notebook, Fill in the blanks*

```
a <- ____ # life expectancy cutoff (<= a)
b <- ____ # infant mortality cutoff (>= b)
c <- ____ # fertility cutoff (>= c)
```

- Combine these conditions with & to create a filter that selects countries that meet all three criteria.

□ *Type this in your R Notebook, Fill in the blanks*

```
needs_screen <- brief_sheet[
  brief_sheet$life_expectancy __ a & # keep low values with which relational operator
  brief_sheet$infant_mortality __ b & # keep high values with which relational operator
  brief_sheet$fertility __ c,      # keep high values with which relational operator
]
```

□ *Verify*

```
nrow(needs_screen)
```

If `nrow(needs_screen) == 0`, adjust the values in a, b, and/or c.

- Explain how `range()` and `sum()` (Task 5) helped you pick cutoffs (a, b, and c) that are reasonable for this dataset.
- Explain how the three conditions are combined to filter the data.

9.7 Task 7 ☒

- | vs %in% (Same Result, Different Syntax)
- Show that filtering the same two regions with | is equivalent to using %in%.

Step 1 Use | to filter for the same two regions you used in Task 3.

```
# □ Type this in your R Notebook, Fill in the blanks
# Use stored region_1 and region_2
# TODO 6-A:
A_or_B <- gapminder[
  gapminder$year == target_year &
  (gapminder$region == region_1 ____ gapminder$region == region_2), # Which operator goes here?
]
```

Step 2 Use %in% to filter for the same two regions.

```
# □ Type this in your R Notebook, Fill in the blanks
# Use the SAME year and regions you used earlier
in_set <- gapminder[
  gapminder$year == target_year &
  gapminder$region ____ c(region_1, region_2), # Which operator goes here?
]
```

Step 3 Compare the number of rows in A_or_B and in_set.

```
# □ Type this in your R Notebook, Fill in the blanks
# TODO 6-B:
nrow(____) ____ nrow(____)
```

□ Briefly compare | and %in%. When is %in% preferable?

9.8 Task 8

Recall from Chapter 8 that which() converts TRUE positions to row indices (numbers). This avoids errors when NA is present and is handy for re-use.

□ Feasibility with which() (Population Window)

□ From needs_screen, keep countries with population between 2 million and 50 million (inclusive) using which().

We'll use e to represent “times 10 to the power of” in R, so 2e6 means 2 million (2×10^6) and 5e7 means 50 million (5×10^7). If these values (2e6 and 5e7) return nothing, you can tweak them to find a solution.

Step 1 Inspect population values first

```
# □ Type this in your R Notebook
range(needs_screen$population, na.rm = TRUE)
```

□ **Tip:** If your range is outside 2e6 to 5e7, you will need to adjust the window.

Step 2 Use which() to build an index of matching rows, then subset with that index to create shortlist.

```
# ☐ Type this in your R Notebook, Fill in the blanks
# TODO 7-A: Build an index of matching rows with which()
idx <- which(needs_screen$population >= 2e6 __ needs_screen$population <= 5e7) # what operators go here?

# ☐ Verify: how many rows matched?
length(idx)
```

☐ If `length(idx) == 0`, widen the window (change `2e6` and/or `5e7`) and re-run Step 1.

Step 3 Use the index `idx` to subset `needs_screen` and create `shortlist`.

```
# ☐ Type this in your R Notebook, Fill in the blanks
# TODO 7-B: Subset rows using idx to create shortlist
shortlist <- needs_screen[____, ____] # subset rows with idx, keep all columns
```

Step 4 Display a compact table of the `shortlist` (choose a few columns to show).

```
# ☐ Type this in your R Notebook
# TODO 7-C: Display a compact table of shortlist (choose a few columns)
shortlist[, c("country", "population", "life_expectancy", "infant_mortality", "fertility")]
```

☐ Explain what `which()` returns, and why might it be useful to store these positions?

9.9 Task 9

☐ Explain the difference between `~`, `!` and `!=`.

9.10 Task 10

☐ **Goal:** Create a well-formatted HTML report for submission.

Step 1: Sweep your Environment (start clean).

Click the ☐ **broom icon** in the Environment pane to clear objects.

Step 2: Run everything top-to-bottom.

Use **Run** → **Run All** to execute every chunk in order.

Step 3: Save and submit.

After your notebook runs without errors, save your notebook (.Rmd) and knit to HTML (.HTML) for submission to Canvas, as directed.

☐ **Tip:** After you knit, look in your course folder for two files with the same name. One file ends in .Rmd and one ends in .html. The HTML usually has a internet browser icon. You can check which is which by double clicking to open them. One opens in R and the other usually opens in a web browser. Upload the .html file. Before submission, check that the uploaded file ends in .html.